

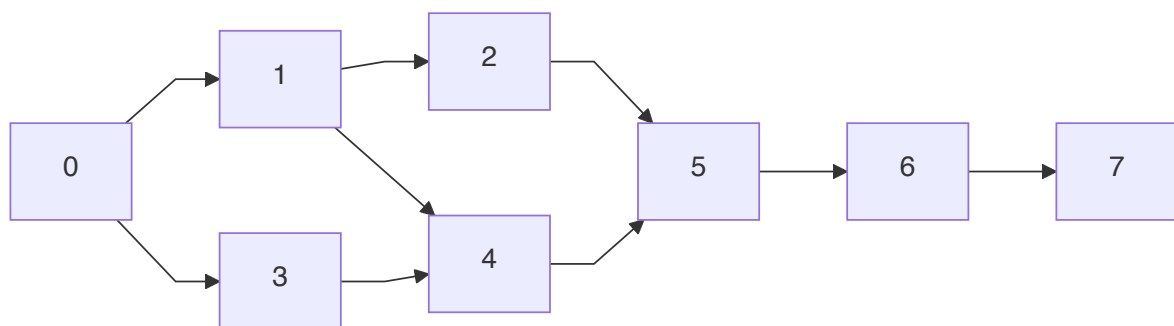
Problem 02: DevOps Pipeline (Topological Sort)

Consider the following software build tasks and their dependencies:

- 1: Linting (No deps)
- 2: Unit Tests (Requires 1)
- 3: Integration Tests (Requires 2)
- 4: Security Scan (Requires 1)
- 5: Build Docker Image (Requires 2 and 4)
- 6: Deploy to Staging (Requires 3 and 5)
- 7: E2E Tests (Requires 6)
- 8: Promote to Prod (Requires 7)

Item a.

Model the dependencies using a DAG (Directed Acyclic Graph).



Adjacency List:

Vertex	Adjacent Vertices
0	1, 3
1	2, 4
2	5
3	4
4	5
5	6
6	7
7	/

Item b.

Provide solutions of **Topological Sort** using DFS and using **BFS**

- **topological DFS solution:** 0, 1, 2, 3, 4, 5, 6, 7
- **topological BFS solution:** 0, 3, 1, 4, 2, 5, 6, 7

Item c.

Assuming that non-dependent tasks can run **in parallel**, what is the minimum number of steps required to complete the pipeline? (Assume each task takes 1 time step).

List the tasks running (including those in parallel) at each step.

Minimum number of steps required to complete the pipeline: 6

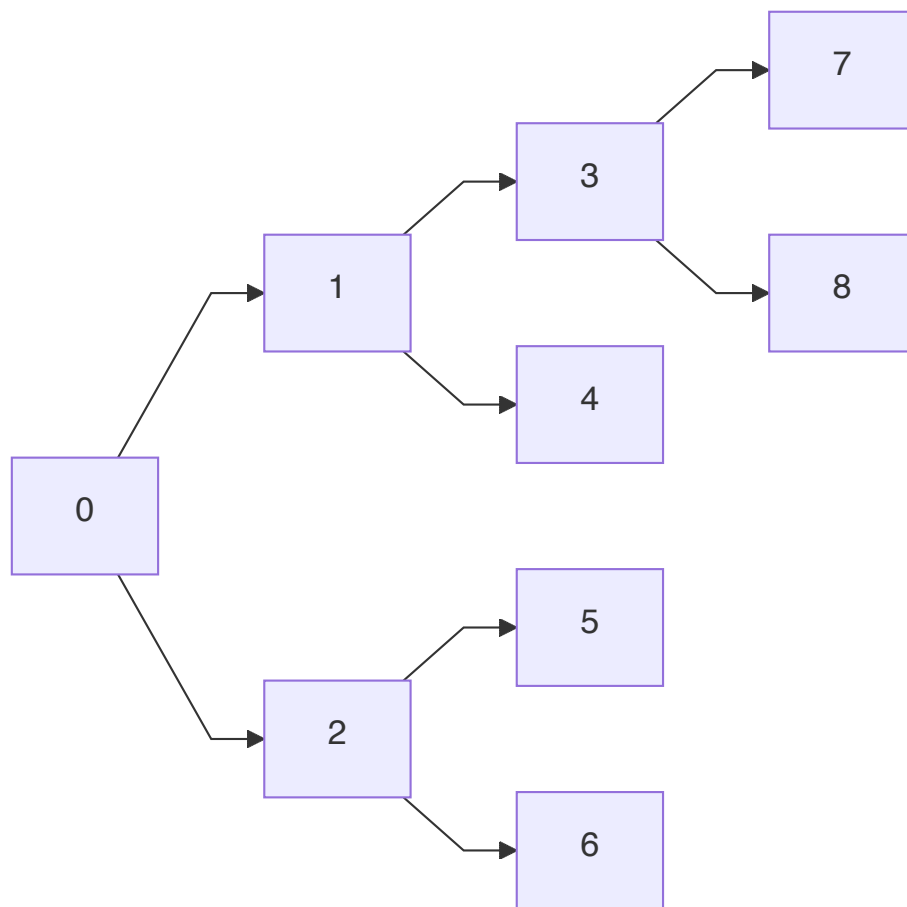
Step	Tasks
1	0
2	1, 3 (since 1 and 3 are non-dependent)
3	2, 4 (since 2 and 4 are non-dependent)
4	5
5	6
6	7

Item d.

Implement the items b. and c. in Java. Provide at least one meaningful test.

See the Java code in [code](#).

Another meaningful test:



- **topological DFS:** 0, 1, 3, 7, 8, 4, 2, 5, 6
- **topological BFS:** 0, 2, 1, 6, 5, 4, 3, 8, 7
- **topological Parallel:** (0), (2, 1), (6, 5, 4, 3), (8, 7)